Assessing the Impact of the Quantitative Easing Shock on Macroeconomic and Financial Variables: A VAR Estimation

Eduardo Beltrão
Sohom Das
Jack Moloney

University of Lausanne

May 22, 2026

Abstract

This study aims to analyze the macroeconomic effects of the European Central Bank's (ECB) quantitative easing (QE) measures taken through the Public Sector Purchase Programme (PSPP). We do so by estimating a VAR(1) model including Euro area PSPP purchases, ECB deposit facility rates, industrial production indices, unemployment, and inflation as endogenous regressors. The impulse response functions show statistically insignificant responses of the endogenous variables to a one-standard-deviation PSPP shock across all horizons. Subsequently, we isolate the clean structural QE shocks from the PSPP purchases through Cholesky decomposition, and study their correlation with key financial variables selected from the ECB's Altavilla dataset.

1 Introduction

1.1 Conceptual Framework

In response to persistently low inflation and interest rates in Europe, as well as the limited effectiveness of conventional monetary policy at the Zero Lower Bound (ZLB), the European Central Bank (ECB) launched an Asset Purchase Programme (APP) in 2015. This programme forms part of the broader Quantitative Easing (QE) framework and involved more than €60 billion in net asset purchases during its first year, and the stock of Eurosystem APP bonds stood at €2402 billion at the end of April 2026. Similar unconventional monetary policies had previously been implemented by other central banks, notably the Federal Reserve in 2011.

Quantitative Easing refers to large-scale asset purchases in which a central bank buys public and private sector bonds in the secondary market. The main objective is to increase liquidity in the financial system and stimulate economic activity when conventional policy instruments, particularly short-term interest rates, are constrained by the ZLB.

Through these purchases, QE increases demand for bonds, which mechanically raises their prices and lowers their yields. Following the announcement of the programme on 22 January 2015, German government bond yields reached historically low levels and even turned negative, reflecting a strong market response to the policy shift.

In this context, Quantitative Easing can be interpreted as a monetary policy shock, defined as an unexpected shift in monetary policy decisions or in the central bank's policy framework. Such shocks may have significant effects on both the financial markets and the broader macroeconomy.

This study aims to assess the impact of this shock on macroeconomic variables, including GDP, unemployment, and inflation, as well as financial variables such as bond yields and exchange rates.

1.2 Literature Review

Theoretical literature identifies several transmission channels through which QE affects the economy. The portfolio rebalancing channel (Tobin, 1969; Gagnon et al., 2011) suggests that central bank purchases reduce the supply of long-term bonds, increasing their prices and lowering yields. The signalling channel (Woodford, 2012) implies that asset purchases communicate a commitment to low interest rates for an extended period. In addition, the credit channel and exchange rate channel operate through improved bank balance sheets and capital flows, respectively.

Empirically, a large body of research has documented the macroeconomic and financial effects of QE. Event-study evidence shows that QE announcements significantly reduce long-term sovereign yields (Gagnon et al., 2011; Krishnamurthy and Vissing-Jorgensen, 2011). For the Euro Area, Altavilla et al. (2015) find that the ECB's APP had substantial effects on sovereign yields and financial conditions. Similarly, Andrade et al. (2016) show that ECB asset purchases led to a significant compression of term premia and improvements in macroeconomic expectations.

Structural empirical approaches, particularly Vector Autoregression (VAR) models, have been widely used to evaluate QE shocks. VAR-based studies generally find that QE leads to lower interest rates, higher asset prices, and expansionary effects on output and inflation (Peersman, 2011; Gambacorta et al., 2014). In the Euro Area context, Boeckx, Dossche and Peersman (2017) show that ECB asset purchase programmes have positive effects on GDP and inflation while reducing sovereign bond yields.

This study contributes to the literature by estimating a VAR model for the Euro Area that incorporates monthly net PSPP purchases as an explicit monetary policy shock. The model jointly examines macroeconomic variables (industrial production, unemployment, and inflation) and financial variables (interest rates and exchange rates), allowing for a comprehensive assessment of the transmission of QE shocks across both real and financial sectors.

2 Data

The data used in this study are collected from several official sources. The analysis focuses on the Public Sector Purchase Programme (PSPP), which represents the main component of the European Central Bank's Asset Purchase Programme (APP). The quantitative easing variable is measured using monthly net purchases at book value under the PSPP, covering the period from March 2015 to January 2022. This series is obtained from the ECB dataset "History of cumulative purchase breakdowns under the APP" available on the European Central Bank's official website.

Macroeconomic variables are sourced primarily from Eurostat. Monthly Industrial Production is used as a proxy for monthly GDP, a standard indicator for assessing business cycle dynamics. In addition, total industrial production for the euro area (2015–2022) and its monthly growth rates are used to capture overall economic activity. Unemployment data is also retrieved from Eurostat.

Inflation is measured using the year-on-year rate of change of the Harmonised Index of Consumer Prices (HICP) for the euro area, also obtained from Eurostat. This measure provides a consistent indicator of consumer price dynamics over time.

Regarding monetary policy, the Deposit Facility rate is used as the key interest rate variable. This series is sourced from the ECB's official database. It is particularly relevant for the period under study, as it served as the main policy instrument during the zero lower bound environment, given that the main refinancing operations rate was constrained by its effective lower bound.

3 Methodology

3.1 Model

We implement a VAR(1) model including several Euro area macroeconomic variables as endogenous regressors in addition to PSP purchases, all on a monthly basis — namely, industrial production (as a proxy for GDP), inflation, unemployment, and the deposit facility rate (the main monetary policy tool and interest rate in the Zero Lower Bound period). Thus, the vector of endogenous variables can be represented by Y_t.

$$Y_t = \begin{bmatrix} PSP_t \\ i_t \\ IP_t \\ u_t \\ \pi_t \end{bmatrix} \tag{1}$$

Here, $PSP_t, i_t, IP_t, u_t,$ and π_t stand for public sector purchases, industrial production, deposit facility rate, unemployment rate, and inflation, respectively, all at time t. Thus, the VAR(1) model estimated can be subsequently specified as follows.

$$Y_t = c + \mathbf{\Phi}_1 Y_{t-1} + \varepsilon_t \tag{2}$$

In the second part of the analysis, we focus on finding correlations between QE shocks and the movement of financial assets. We identify structural QE shocks from PSPP purchases by placing PSP_t first in the Cholesky ordering. This ensures that the residual ε_{1t} from the PSP_t equation in the above VAR(1) model, as shown below, is the clean structural QE shock. We place interest rates second as we expect the main monetary policy instrument of the ECB to be reactive to quantitative easing measures taken. The other variables are placed afterwards, as they are slow-moving macroeconomic variables.

3.2 Non-stationarity Tests

We use year-on-year monthly growth rates in the industrial production index (calculated relative to 2021 levels = 100), which makes the IP_t variable stationary by construction. However, conducting non-stationarity tests on this variable showed the KPSS and Phillips-Perron (PP) tests to confirm stationarity, while the ADF test failed to reject the null. We infer this to be due to the small sample size, which spans over seven years, as that is the duration of the PSPP purchases programme. As two out of three tests confirm stationarity and the year-on-year transformation is stationary by construction, we treat IP_t as stationary and proceed.

KPSS and ADF tests also revealed the need to first-difference $PSP_t, i_t, u_t,$ and π_t. Carrying out the ADF and KPSS tests again on the first-differenced variables confirmed stationarity for PSP_t and u_t. However, as the deposit facility rate and inflation both remained non-stationary, we followed up on these variables with Phillips-Perron tests and visual inspection. The graphs used for visual inspection of these two variables are provided in Appendix A. The graph for first-differences of inflation over time showed a seemingly stationary process, with the Phillips-Perron test confirming this. For i_t, the graph was not clearly stationary, with rates resting at zero for long periods of time. However, given that i_t also passed the PP test and its lengthy flat periods are a product of ECB policy, we moved forward with our analysis using first-differences, which will be referred to henceforth as the first-differences specification. We interpret the non-stationarity results as being driven by the small sample size, which prevents the tests from observing the variables' mean reversion cycles. Furthermore, given that interest rates and inflation are typically considered stationary, we also re-estimate the VAR(1) model using the original monthly *levels* of inflation and the deposit facility rate (not first-differenced), which will be referred to henceforth as the levels specification.

3.3 Lag Selection

Lag length selection via information criteria resulted in AIC suggesting 12 lags and FPE suggesting six lags, while both HQ and SC suggested two lags. Given that we consider a maximum lag length of 12 due to the monthly frequency of our data, and our small sample size of 92 observations, we interpreted AIC's suggestion of 12 lags to be due to overfitting. As both HQ and SC suggest a lag length of two, we first estimate a VAR(2). However, this yielded a root of 1.1766, exceeding one. We thus subsequently reduced the lag length to one, which produces a stable VAR.

4 Empirical Results

In this section, we present the empirical results from our methodology detailed above. This includes the impulse response functions (IRFs) from the estimation of the VAR(1) model, the cleanly recovered structural QE shocks, and the autocorrelation tests conducted on them, as well as the correlation analyses with financial variables done thereafter.

4.1 Impulse Response Functions (IRFs)

The VAR finds no statistically significant macro transmission of ECB QE shocks through standard channels. The IRFs show that the responses of the selected macro variables to an unexpected QE shock through PSPP purchases are insignificant at the 95% confidence interval level. This is consistent with the view that QE transmission operates primarily through financial market channels, such as bond yields and asset prices, rather than directly through aggregate macro variables. We also estimate the own-IRF of an unexpected QE shock on PSPP purchases, which is shown below. This demonstrates the identified QE shock through Cholesky decomposition to be transitory, as the impulse response of PSPP purchases decays to zero within five months. Thus, the extracted shock represents genuine month-to-month surprises in ECB purchase decisions rather than persistent systematic deviations. The plots showcasing the other IRFs in both the first-differences and levels specifications are provided in appendices B and C, respectively.

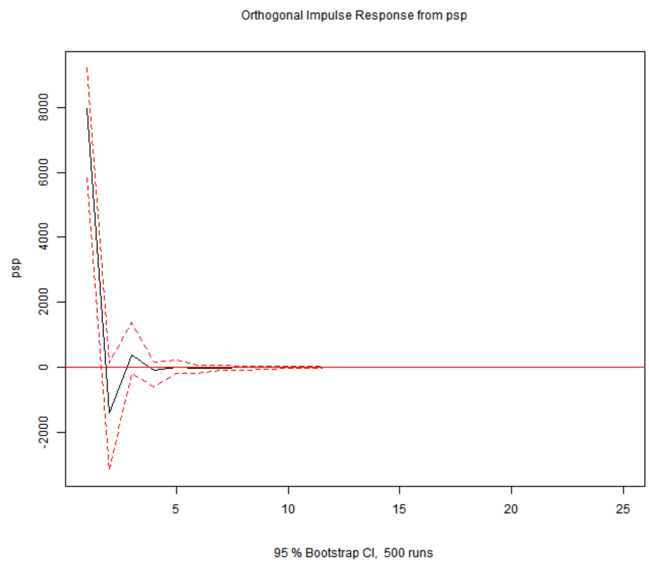


Figure 1: Impulse response of PSPP purchases to QE shock recovered through Cholesky decomposition with first-differenced inflation and interest rates.

4.2 QE Structural Shocks and Autocorrelation Tests

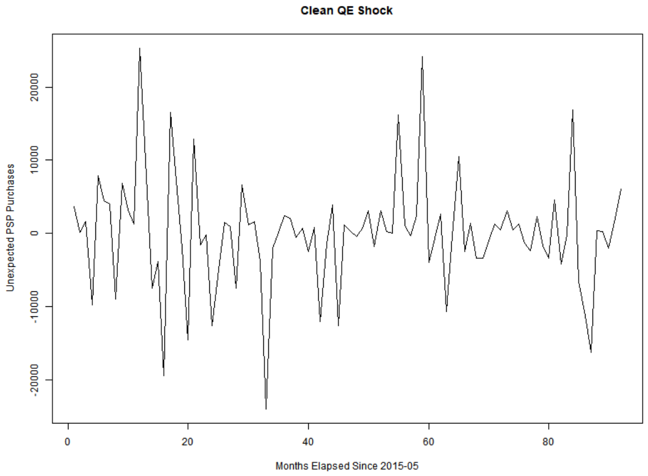


Figure 2: Extracted structural QE shock series through Cholesky decomposition with first-differenced inflation and interest rates.

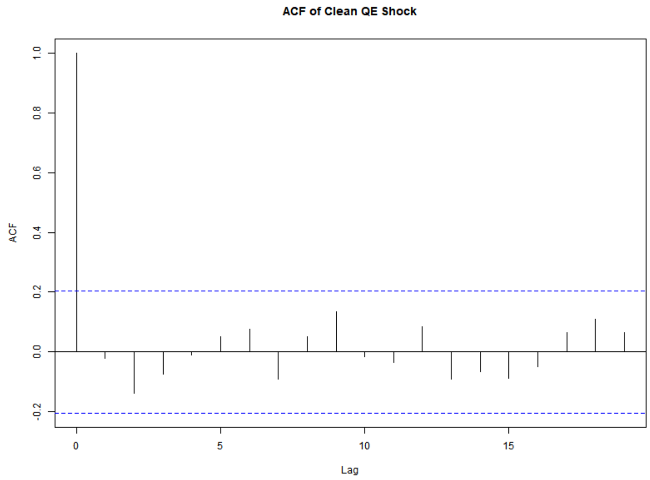


Figure 3: Autocorrelation Function (ACF) of Extracted QE shock series in the first-differences specification.

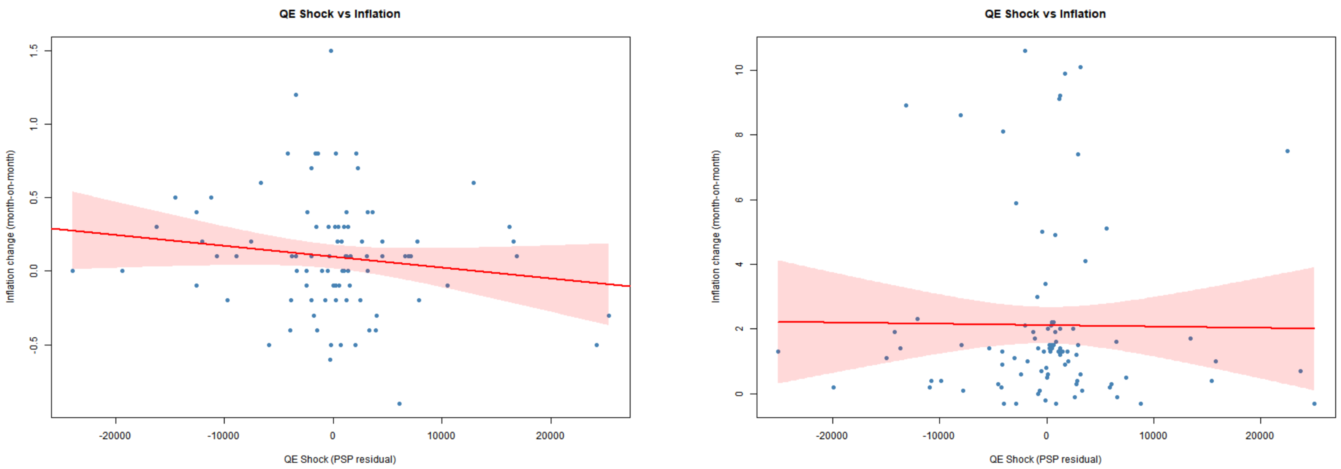
The clean rescued QE structural shock over time is presented above in Figure 2. The series fluctuates around zero throughout, consistent with what is expected of white noise. Positive values indicate months where ECB asset purchases exceeded model predictions, while negative values indicate months where purchases fell short of expectations. To validate the identification strategy, we test the extracted QE shock series for serial autocorrelation, with Figure 3 showing the autocorrelation function (ACF). All bars lie within the 95% confidence bands at every lag, indicating no statistically significant autocorrelation. The partial autocorrelation function (PACF) further confirms this result, and is provided in Appendix D. We recover similar white noise results for the QE Shock retrieved from the levels specification as well, provided in Appendix E. Together, these results show that the extracted shock series is white noise, validating the Cholesky identification strategy.

4.3 Correlation Analysis

To assess the financial market transmission of QE shocks, we plot the VAR-extracted PSP shock against key financial variables from the Euro Area Monetary Policy Event-Study Database (EA-MPD). Specifically, we examine the co-movement between the QE shock and four financial variables measured on ECB meeting days: the 2-year OIS rate, the German 10-year Bund yield (DE10Y), the Italian 10-year sovereign yield (IT10Y), and the EUR/USD exchange rate. All financial variables are taken from the Press Conference Window of the EA-MPD, capturing intraday price changes in the two-hour window following ECB announcements, and aggregated to monthly frequency by summing changes across all meetings within a given month.

As a benchmark, Figure 4 plots the PSP shock against monthly HICP inflation under both specifications. Panel (a) uses the first-differenced QE shock extracted from the baseline VAR, yielding a near-zero correlation of $r = -0.147$. Panel (b) re-estimates the clean QE shock using a levels specification of the VAR and plots it against the HICP inflation level, yielding $r = -0.012$. Both panels confirm that raw PSPP purchase volumes have no discernible relationship with inflation outcomes, consistent with the insignificant IRF findings reported above.

These results are robust to the choice of variable transformation, as the near-identical correlations across both specifications confirm. The finding is consistent with a well-established result in the monetary policy literature: the macroeconomic effects of central bank asset purchases are driven by the *surprise* component of announcements, not by total purchase volumes. Since PSPP targets were largely pre-announced and anticipated by markets, the VAR-extracted shock conflates anticipated and unanticipated components, attenuating any detectable transmission to inflation.



(a) First Differences

(b) Levels

Figure 4: QE Shock vs HICP Inflation. Panel (a) plots the first-differenced QE shock against month-on-month inflation change. Panel (b) plots the shock in levels. Both panels confirm a near-zero correlation, suggesting that raw PSPP purchase volumes have no discernible relationship with inflation outcomes regardless of the transformation applied.

	First Differences		Levels	
Variable	**Correlation**	**T-Stat**	**Correlation**	**T-Stat**
OIS 2Y	-0.094	-0.723	-0.096	-0.744
German Bund 10Y	-0.173	-1.352	-0.179	-1.401
Italian 10Y	-0.012	-0.093	-0.017	-0.133
EUR/USD	-0.064	-0.489	-0.069	-0.535
HICP Inflation	-0.147	-1.409	-0.012	-0.116

Table 1: Pairwise Correlations between QE Shock and Financial Variables. *Note:* QE shock is the VAR residual from the PSPP equation. Financial variables are aggregated monthly sums of intraday changes from the EA-MPD Press Conference Window. None of the correlations are statistically significant at conventional levels.

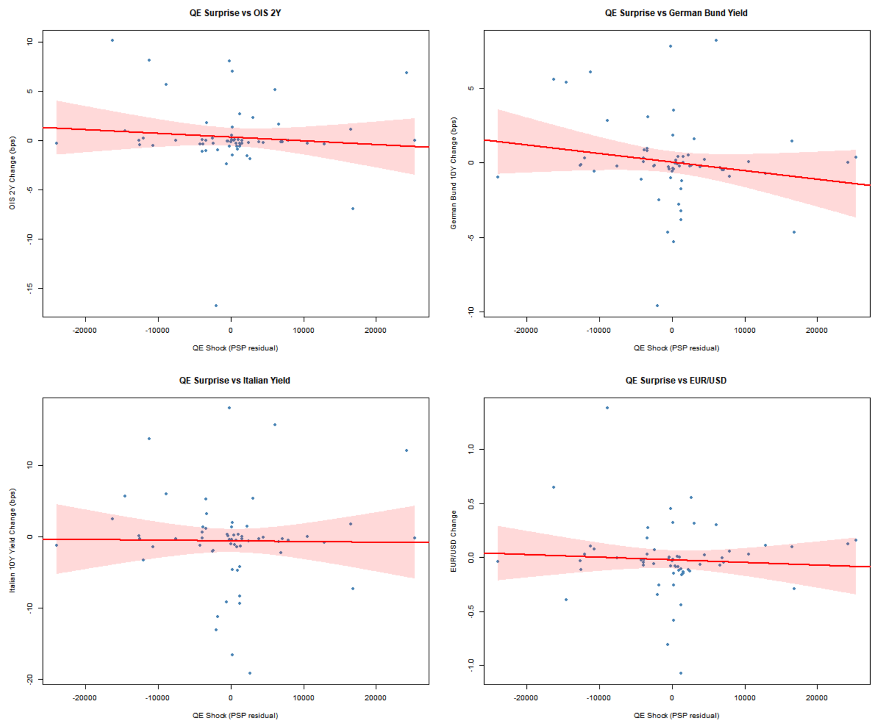


Figure 5: QE Shock vs Financial Variables (EA-MPD Press Conference Window). Scatter plots show the relationship between the VAR-extracted PSP shock and OIS 2Y, German Bund 10Y, Italian 10Y sovereign yield, and EUR/USD exchange rate. Red lines indicate OLS fit. All correlations are weak and near-zero, consistent with the hypothesis that anticipated purchase volumes carry limited surprise information.

Figure 5 plots the PSP shock against the four EA-MPD financial variables. Table 1 reports the pairwise correlations under both specifications. The results show weak and near-zero correlations across all four financial variables, with the strongest relationships observed for the German Bund yield ($r = -0.173$ in first differences, $r = -0.179$ in levels) and OIS 2Y ($r = -0.094$ in first differences, $r = -0.096$ in levels). The near-identical correlations across both specifications further confirm robustness to transformation choice. These weak correlations are consistent with the IRF findings and further confirm that raw PSPP purchase volumes — which are largely anticipated by markets — carry limited information for financial market outcomes. This motivates the use of high-frequency surprise measures, such as those provided directly by the EA-MPD, as a cleaner identification strategy for future research.

5 Conclusion

The empirical results of this study point to two main findings that together shed light on the transmission mechanism of ECB Quantitative Easing.

Macroeconomic transmission is weak. The impulse response functions show statistically insignificant responses of industrial production, unemployment, inflation, and interest rates to a one-standard-deviation PSPP shock across all horizons. This is further corroborated by the Granger causality test, which fails to reject the null hypothesis that PSPP purchases do not Granger-cause macroeconomic outcomes ($F = 0.695$, $p = 0.596$). The correlation between the PSP shock and HICP inflation is near-zero ($r = -0.147$), confirming that raw purchase volumes carry no meaningful predictive content for price dynamics.

Financial market transmission is limited under raw purchase volumes. The scatter plots and correlation table reveal weak co-movement between the VAR-extracted PSP shock and financial variables from the EA-MPD, with correlations ranging from $r = -0.012$ for Italian sovereign yields to $r = -0.173$ for German Bund yields. All correlations are negative, consistent with the expected direction — larger purchase volumes should compress yields and weaken the euro — but the relationships are too noisy to be economically meaningful.

Identification is the key limitation. These findings are consistent with a well-established result in the monetary policy literature: the macroeconomic and financial effects of central bank actions are driven by the *surprise* component of policy announcements, not by the total volume of purchases. Since PSPP purchase targets were largely pre-announced and anticipated by markets, the VAR-extracted shock conflates anticipated and unanticipated components, attenuating any detectable transmission. This motivates the use of high-frequency identified surprise measures — such as those provided directly by the EA-MPD — as a more appropriate shock variable, which we leave for future research.

Taken together, these results do not imply that ECB QE was ineffective. Rather, they highlight the importance of shock identification in empirical monetary policy research: the appropriate measure of a QE shock is not the volume of purchases but the unexpected component thereof, as revealed by high-frequency asset price movements around announcement dates.

References

[1] Tobin, J. (1969). *A General Equilibrium Approach to Monetary Theory*. Journal of Money, Credit and Banking, 1(1), 15–29.

[2] Gagnon, J., Raskin, M., Remache, J., & Sack, B. (2011). *The Financial Market Effects of the Federal Reserve's Large-Scale Asset Purchases*. International Journal of Central Banking, 7(1), 3–43.

[3] Woodford, M. (2012). *Methods of Policy Accommodation at the Interest-Rate Lower Bound*. In *The Changing Policy Landscape* (Proceedings of the Federal Reserve Bank of Kansas City Economic Symposium, Jackson Hole).

[4] Krishnamurthy, A., & Vissing-Jorgensen, A. (2011). *The Effects of Quantitative Easing on Interest Rates: Channels and Implications for Policy*. Brookings Papers on Economic Activity, 43(2), 215–287.

[5] Altavilla, C., Carboni, G., & Motto, R. (2015). *Asset Purchase Programmes and Financial Markets: Lessons from the Euro Area*. ECB Working Paper No. 1864.

[6] Altavilla, C., Brugnolini, L., Gürkaynak, R. S., Motto, R., & Ragusa, G. (2019). *The Euro Area Monetary Policy Event-Study Database*.

[7] Andrade, P., Breckenfelder, J., De Fiore, F., Karadi, P., & Tristani, O. (2016). *The ECB's Asset Purchase Programme: An Early Assessment*. ECB Working Paper No. 1956.

[8] Peersman, G. (2011). *Macroeconomic Effects of Unconventional Monetary Policy in the Euro Area*. CESifo Working Paper No. 3589.

[9] Gambacorta, L., Hofmann, B., & Peersman, G. (2014). *The Effectiveness of Unconventional Monetary Policy at the Zero Lower Bound: A Cross-Country Analysis*. Journal of Money, Credit and Banking, 46(4), 615–642.

[10] Boeckx, J., Dossche, M., & Peersman, G. (2017). *Effectiveness and Transmission of the ECB's Balance Sheet Policies*. International Journal of Central Banking, 13(1), 297–333.

[11] Kuttner, K. N. (2001). *Monetary Policy Surprises and Interest Rates: Evidence from the Fed Funds Futures Market*. Journal of Monetary Economics, 47(3), 523–544.

[12] Gürkaynak, R. S., Sack, B., & Swanson, E. (2005). *Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements*. International Journal of Central Banking, 1(1), 55–93.

[13] European Central Bank (ECB). *History of cumulative purchase breakdowns under the APP (PSPP dataset)*. Available at the ECB Statistical Data Warehouse.

[14] European Central Bank (ECB). *Deposit Facility Rate*. ECB Statistical Data Warehouse.

[15] European Central Bank (ECB). *Euro Area Unemployment Rate*. ECB Statistical Data Warehouse.

[16] Eurostat. *Industrial Production Index (Euro Area)*.

[17] Eurostat. *Harmonised Index of Consumer Prices (HICP), Euro Area*.

Appendix

Appendix A

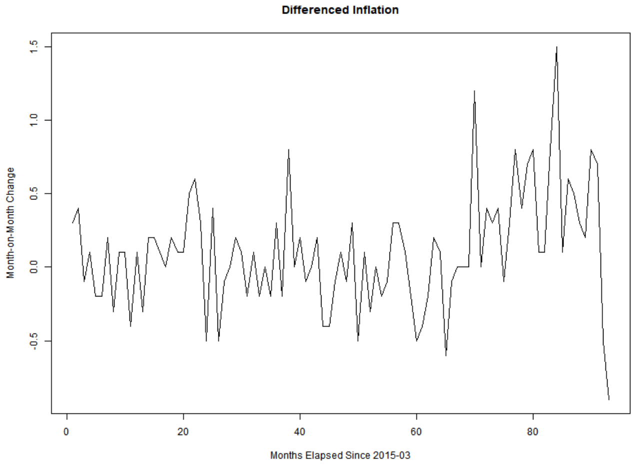


Figure 6: Differenced Inflation over time.

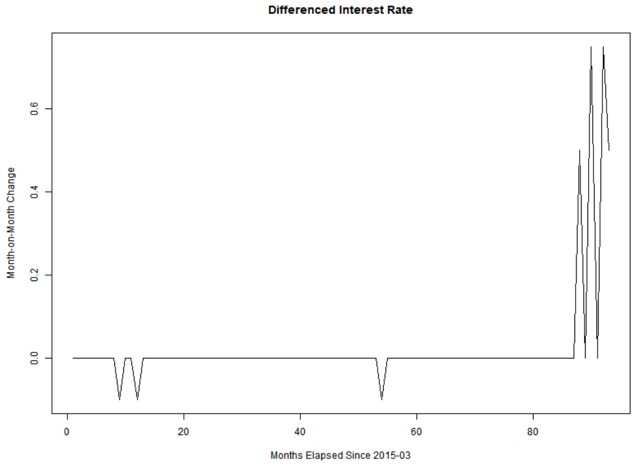


Figure 7: Differenced Interest Rate over time.

Appendix B

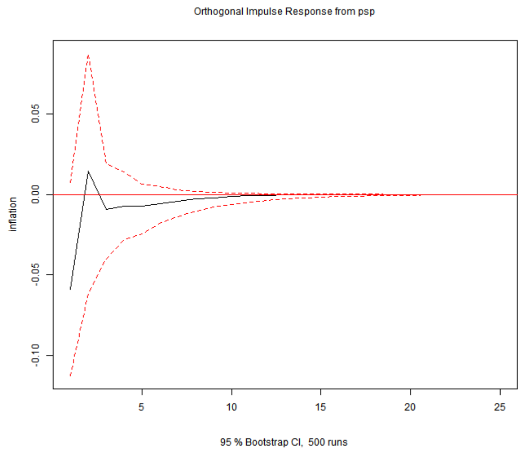


Figure 8: Impulse Response of Inflation to QE Shock (first-differenced inflation and interest)

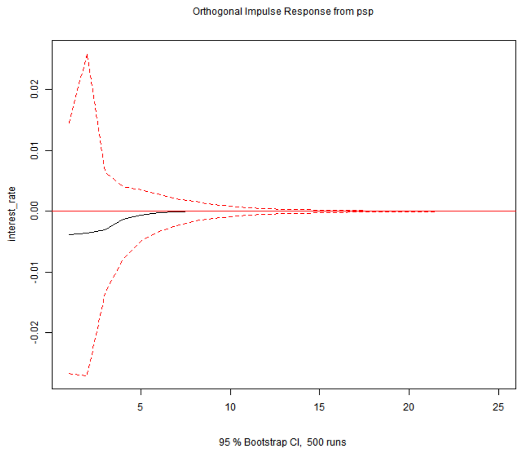


Figure 9: Impulse Response of Interest Rate to QE Shock (first-differenced inflation and interest)

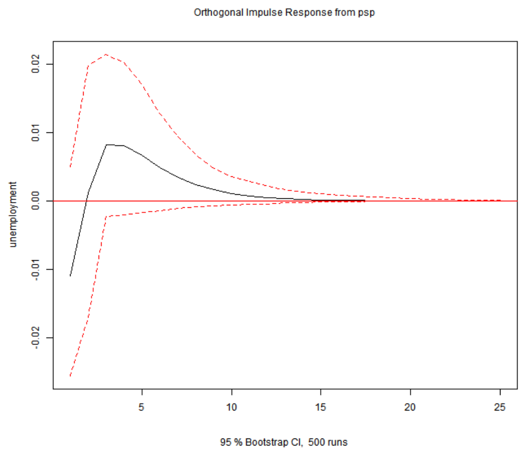


Figure 10: Impulse Response of Unemployment to QE Shock (first-differenced inflation and interest)

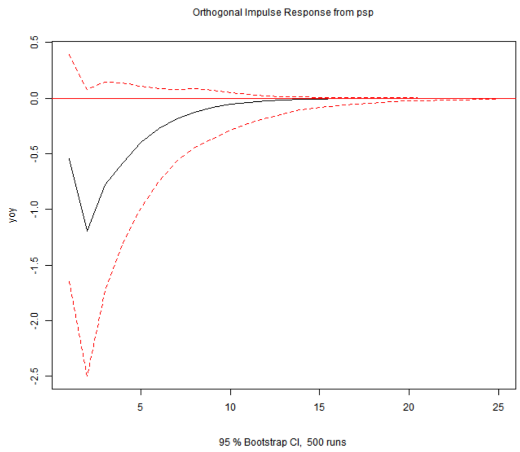


Figure 11: Impulse Response of Year-on-Year Industrial Production to QE Shock (first-differenced inflation and interest)

Appendix C

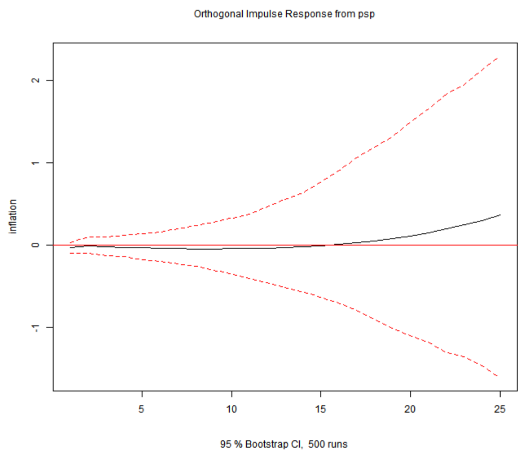


Figure 12: Impulse Response of Inflation to QE Shock (monthly levels of inflation and interest)

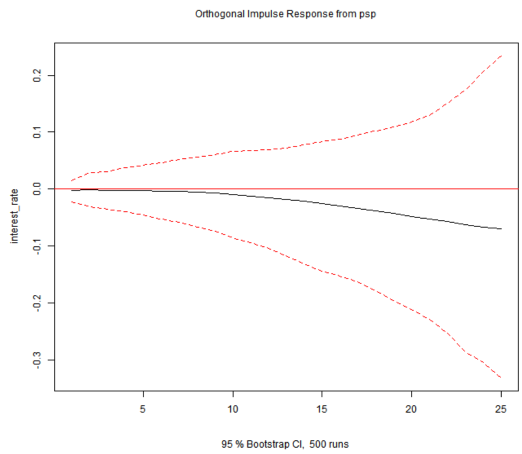


Figure 13: Impulse Response of Interest Rate to QE Shock (monthly levels of inflation and interest)

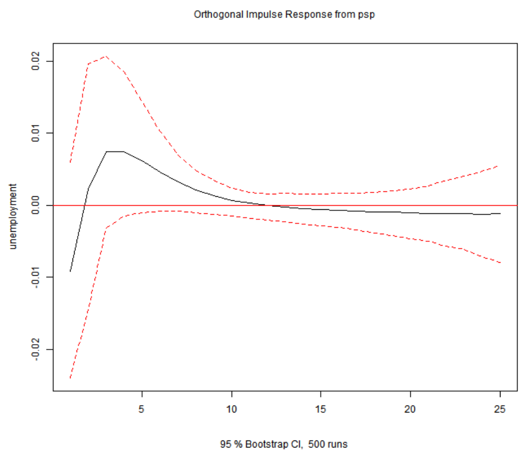


Figure 14: Impulse Response of Unemployment to QE Shock (monthly levels of inflation and interest)

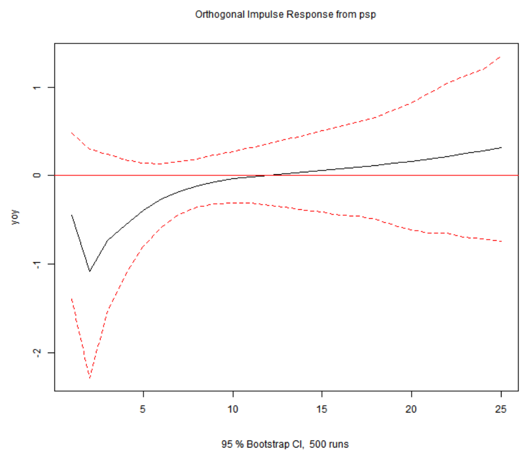


Figure 15: Impulse Response of Year-on-Year Industrial Production to QE Shock (monthly levels of inflation and interest)

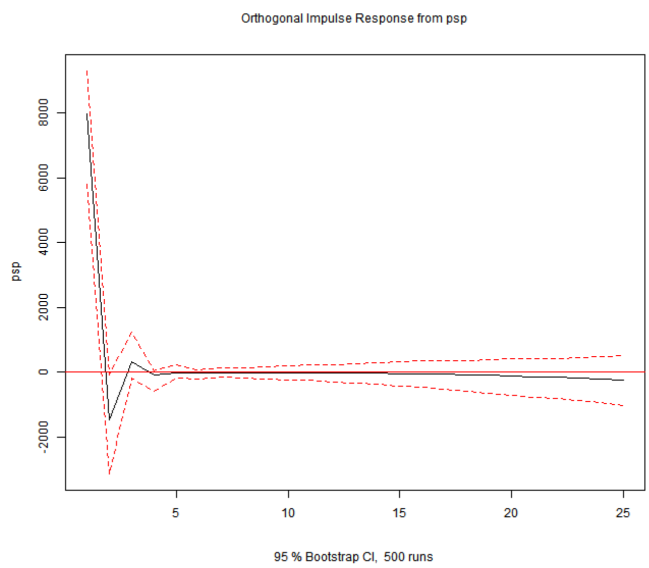


Figure 16: Impulse response of PSPP purchases to QE shock recovered through Cholesky decomposition with monthly levels of inflation and interest rates.

Appendix D

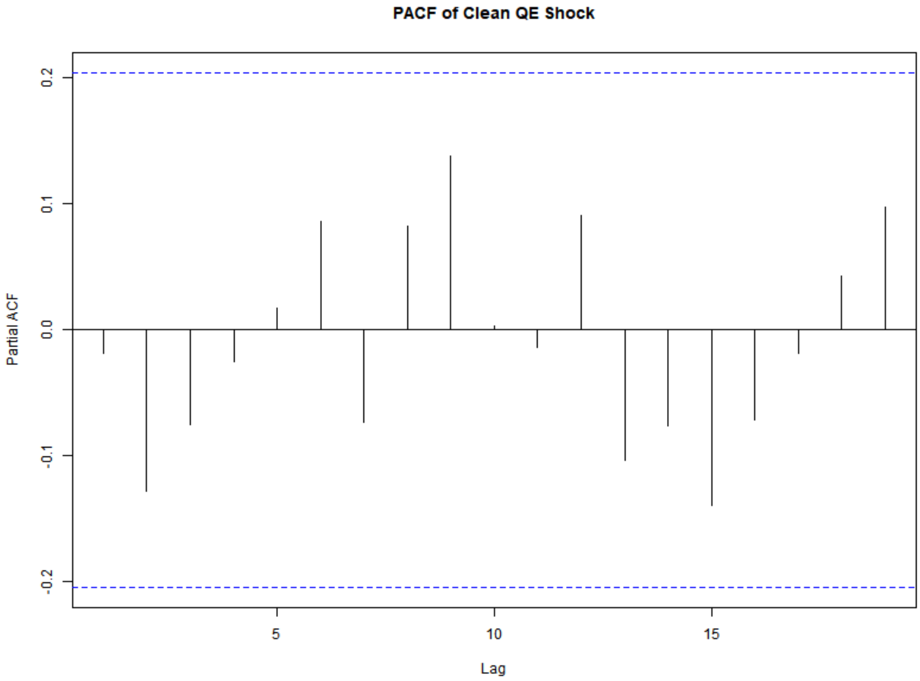


Figure 17: PACF of QE shock series extracted through Cholesky decomposition in the first-differences specification

Appendix E

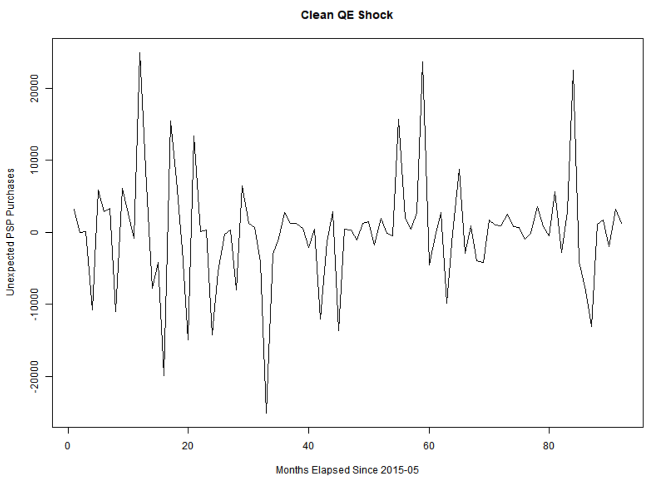


Figure 18: Extracted structural QE shock series through Cholesky decomposition with monthly levels of inflation and interest rates.

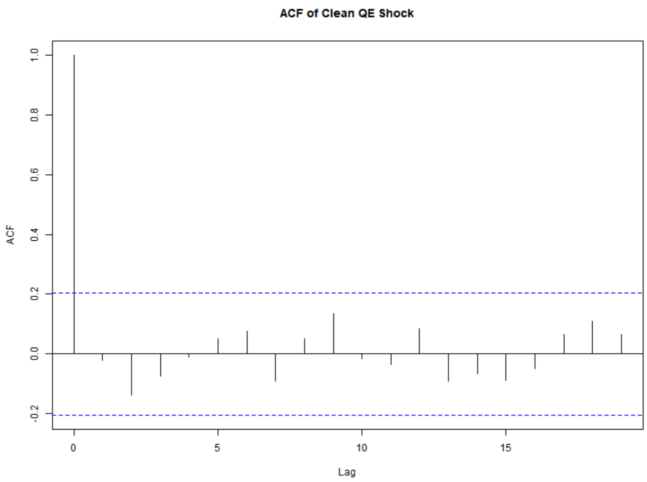


Figure 19: ACF of extracted QE shock series in the levels specification.

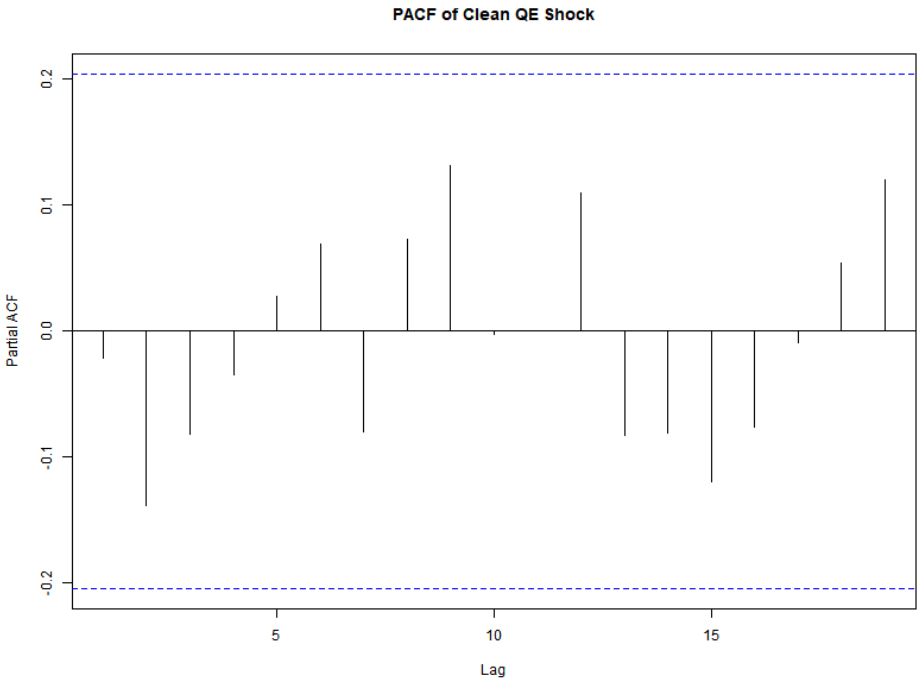


Figure 20: PACF of QE shock series extracted through Cholesky decomposition in the levels specification